Name _____ Period _____ Date _____

4-1 Lesson Quiz: Inverse Variation and the Reciprocal Function short form

3. Three students can wash a car in 16 minutes. If the time varies inversely with the number of students washing the car, how many minutes will it take two students to complete that same job? *(bonus)*

Ⓐ 20 Ⓑ 22 Ⓒ 24 Ⓓ 26

Show your work. Equation: $t =$ _____ $k =$ _____

4. What is the equation of the horizontal asymptote of the graph of $y = \frac{1}{x+5} - 4$?

Horizontal asymptote: _____

Explain how you know from the equation: _____

5. The graph of $y = \frac{1}{x}$ is translated 3 units up and 2 units to the left. What is an equation of the translated graph?

Ⓐ $y = \frac{1}{x+2} + 3$ Ⓑ $y = \frac{1}{x-2} + 3$

Ⓒ $y = \frac{1}{x-3} + 2$ Ⓓ $y = \frac{1}{x+3} - 2$

Explain your choice: which part of the equation shows the move left, and which shows the move up?

Scoring: Question 4 and Question 5 are 10 points each. Question 3 is a bonus of up to 5 points. Total out of 20.

4-1 Lesson Quiz, short form: answer key

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3. C, 24 minutes. With n students, $t = \frac{k}{n}$; $16 = \frac{k}{3}$, so $k = 48$; $t = \frac{48}{2} = 24$. Savvas quiz item 3, DOK 1, Realize skill F74. Taught by Example 3 (cut from the fast packet); the packet's Extension (Practice 20, the pump) is the same two steps; IXL PNY. **Bonus 5:** 5 for C with $k = 48$ shown; 3 for C alone.

4. $y = -4$**.** $y = \frac{1}{x+5} - 4 = \frac{1}{x-(-5)} + (-4)$ is $y = \frac{1}{x}$ translated 5 left and 4 down, so $h = -5$, $k = -4$ and the horizontal asymptote is $y = k = -4$. Savvas quiz item 4, DOK 2, Realize skill F74. Taught by Example 5 and Try It 5 ($y = \frac{1}{x+2} - 4$ has the same k); IXL ULD. **10 points:** 8 for $y = -4$ (6 for -4 written without $y =$), 2 for the explanation (the -4 added outside the fraction moves the asymptote $y = 0$ down 4).

5. A, $y = \dfrac{1}{x+2} + 3$**.** Left 2 means $h = -2$, so $x - h = x + 2$; up 3 means $k = 3$, added outside. Savvas quiz item 5, DOK 2, Realize skill F74. Taught by Example 5 and the packet's Practice (a parallel with different numbers); IXL F6J. B has the wrong sign for a move left; C and D swap the horizontal and vertical moves. **10 points:** 8 for A, 2 for the explanation (+2 inside the denominator is the move left, +3 outside is the move up).

Items left off this form (available for a retake or Form B; both are inverse-variation tables taught by Examples 1-3, which the fast packet cuts): Savvas item 1, the missing value p in the table x: 3, 10, 15, 30; y: 5, p, 1, 0.5 (answer $p = 1.5$, DOK 1, F74); Savvas item 2, the equation for the table x: -3, -1, $\frac{1}{2}$, $\frac{2}{3}$; y: 4, 12, -24, -18 (answer D, $y = \frac{-12}{x}$, DOK 2, F74). The TE's Lesson Quiz has no graphs, so nothing was redrawn.

IXL replacement: SmartScore 70 or above on ULD, F6J and PNY replaces this quiz with 20/20.